



Synapse

Bridging the Biological Interpretation Gap

Multimodal Biological Design & Discovery

Meet Our Team



**EYA
CHIKHAOUI**



**MOUHAMED
GHARSALLAH**



**BALKIS
MAHJOUBI**



**MOHAMED AZIZ
MASGHOUNI**



TABLE OF CONTENT

01

PROBLEM STATEMENT

The computational bottleneck limiting modern genomics

02

YOUR SOLUTION

Geometric retrieval paradigm for DNA-protein mapping

03

ARCHITECTURE

Dual-transformer engine with unified vector space

04

HOW IT WORKS

Geometric retrieval paradigm for DNA-protein mapping

05

DEMO & RESULTS

Validation metrics and real-world performance benchmarks

06

IMPACT & VALUE

Transforming drug discovery and clinical genomics

01

PROBLEM STATEMENT

THE GLOBAL ISSUE

Sequencing is fast ($O(N)$),
Understanding is slow ($O(N^3)$)

THE REAL-WORLD CONTEXT

Amplifies DNA for analysis.

THE "PHYSICS BARRIER

High-Throughput screening
is impossible with current
tools

Case Study

Optimization pipelines (like those
at InstaDeep) are limited by
compute latency

02

YOUR SOLUTION

PARADIGM SHIFT

Simulation" to "Geometric Retrieval

MECHANISM

Mapping DNA and Proteins
to the same vector space

CORE TECH

Shared Latent Space

Result

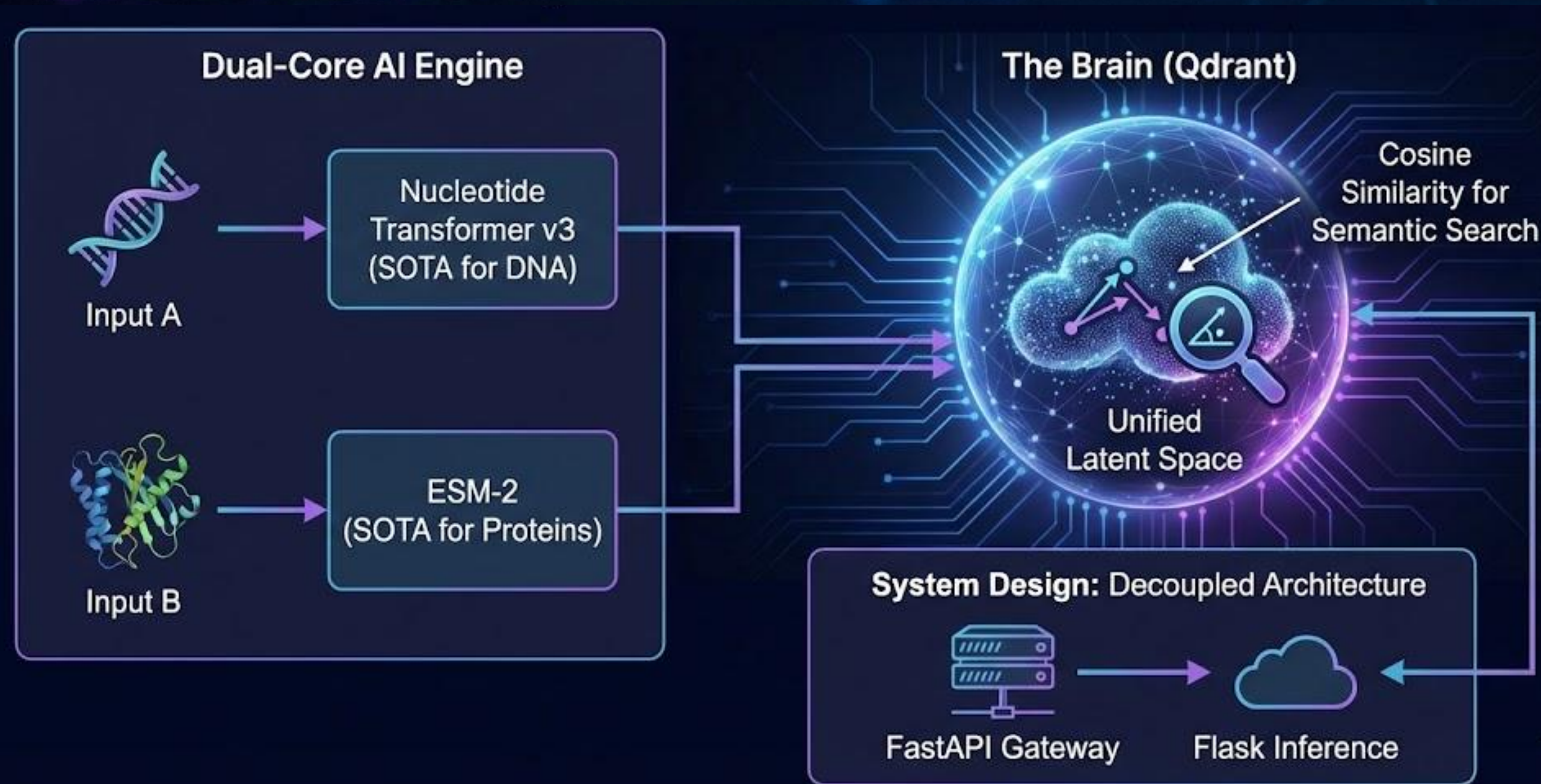
Real-time semantic alignment





03

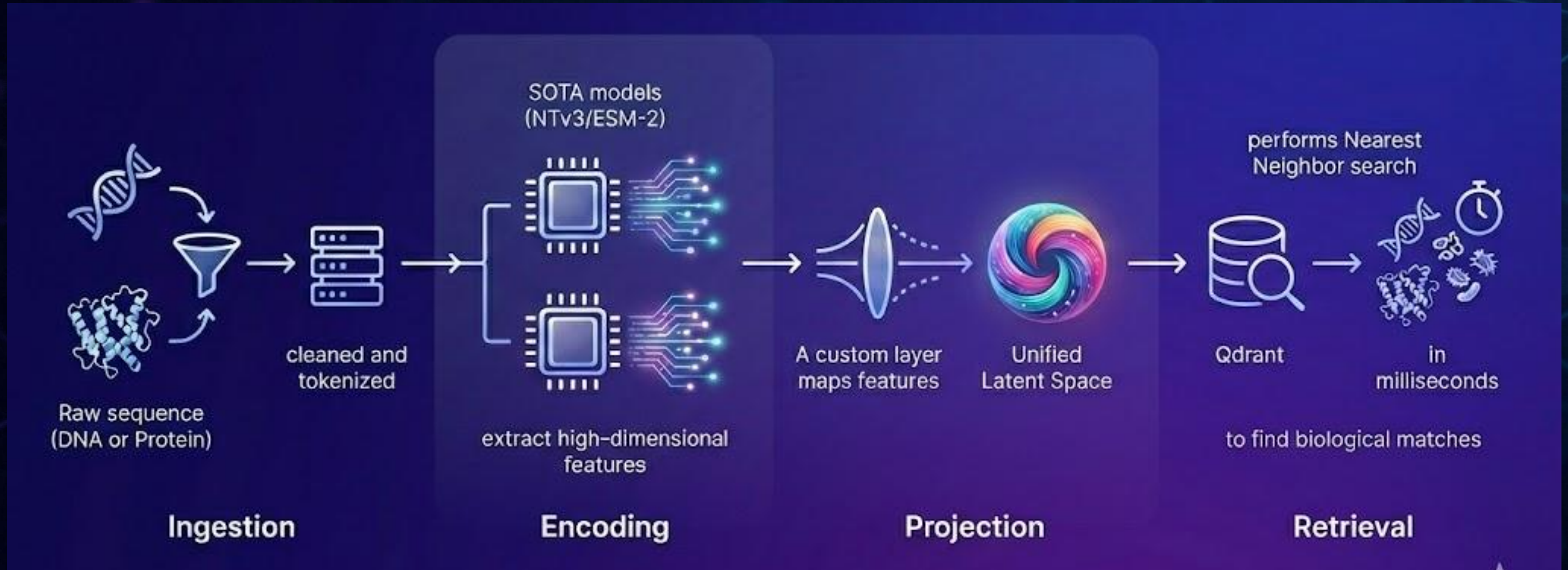
ARCHITECTURE - CORE FEATURES





04

HOW IT WORKS





KEY METRICS & PERFORMANCE

44.1%

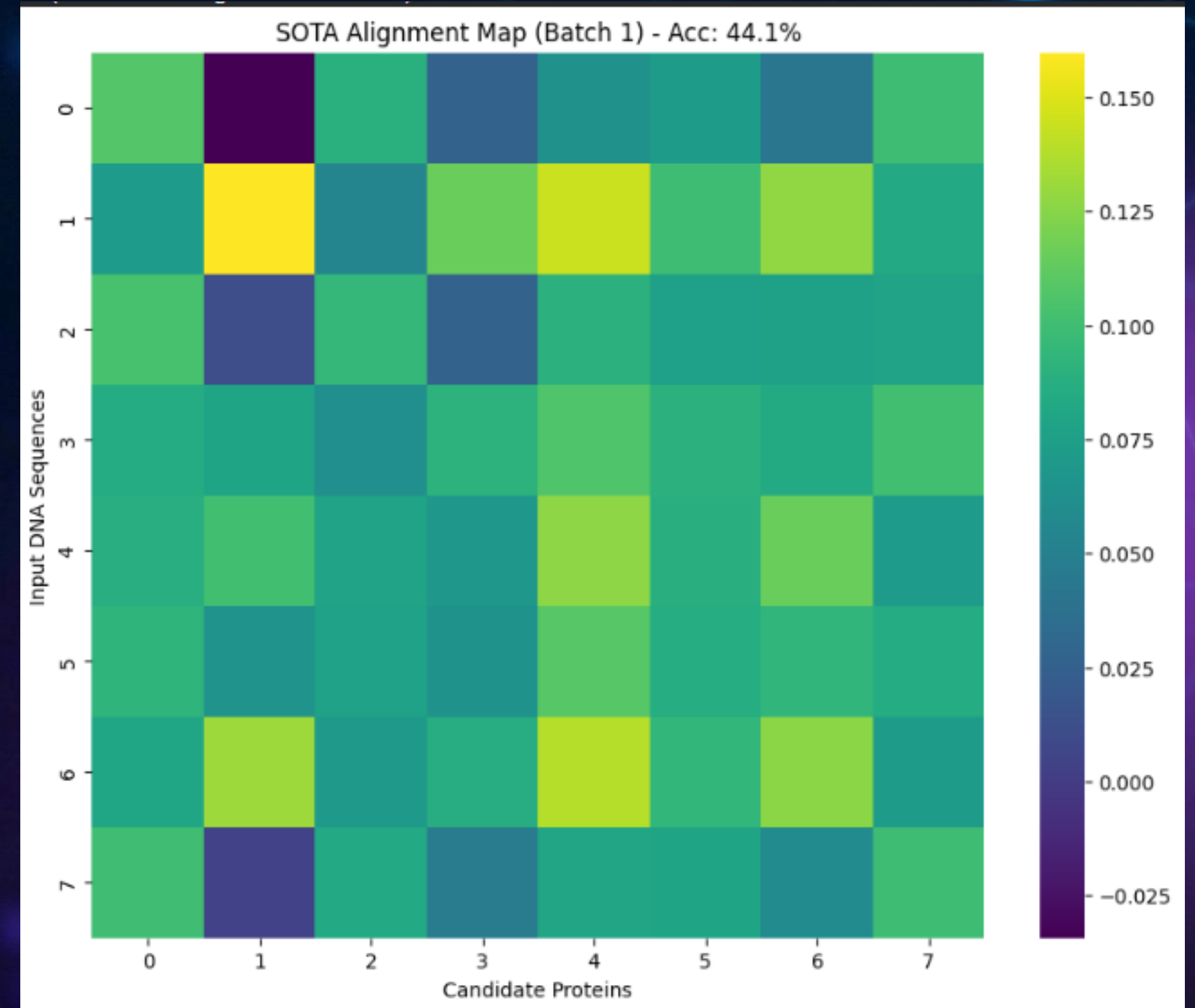
Zero-Shot Accuracy

On unseen genomic data, significantly exceeding <1% random baseline.

<60ms

Inference Latency

Enabling real-time reinforcement learning feedback loops.





VISUALS & TRAINING STABILITY

- Loss curves show steady optimization over 5 Epochs.
- Model successfully minimized the contrastive loss between DNA and Protein vectors.

```
... 🔥 STARTING SOTA TRAINING (GOD MODE)
    Physical Batch: 8 (Safe)
    Accumulation: 128 steps
    Effective Batch: 1024 (Target: 1024 - Google Scale)
-----
/tmp/ipython-input-1070334713.py:29: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use `torch.amp.GradScaler('cuda', args...)` instead.
    scaler = GradScaler()

Epoch 1: 100% ██████████ 750/750 [12:19<00:00, 1.18it/s, acc=25%, loss=2.0510]
/tmp/ipython-input-1070334713.py:47: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
    with autocast():
    ★ New Best Model! (Loss: 2.0720) -> Saving...
Epoch 1 Completed.

Epoch 2: 100% ██████████ 750/750 [12:22<00:00, 1.32it/s, acc=12%, loss=2.0483]
    ★ New Best Model! (Loss: 2.0492) -> Saving...
Epoch 2 Completed.

Epoch 3: 100% ██████████ 750/750 [12:23<00:00, 1.19it/s, acc=25%, loss=1.9924]
    ★ New Best Model! (Loss: 1.9994) -> Saving...
Epoch 3 Completed.

Epoch 4: 100% ██████████ 750/750 [12:21<00:00, 1.25it/s, acc=38%, loss=1.8003]
    ★ New Best Model! (Loss: 1.8887) -> Saving...
Epoch 4 Completed.

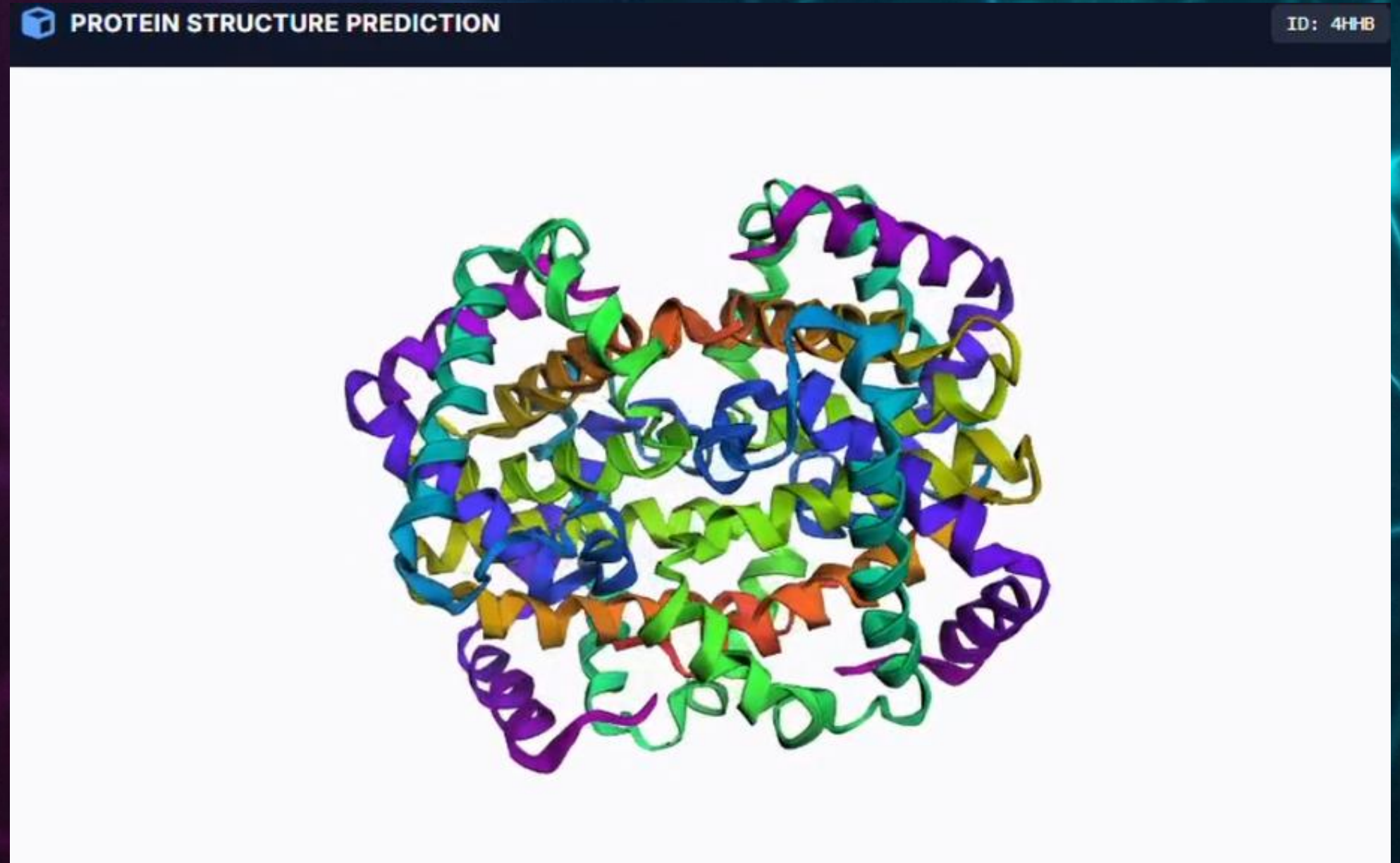
Epoch 5: 100% ██████████ 750/750 [12:26<00:00, 1.35it/s, acc=62%, loss=1.4988]
    ★ New Best Model! (Loss: 1.6932) -> Saving...
Epoch 5 Completed.
📁 Titan Training Complete.
```




ATHER FEATURES

From Code to Structure

Instantly renders the **3D Protein Structure** associated with the retrieved **DNA/Protein sequence**.





IMPACT & VALUE

THE DIMENSIONALITY GAP

- **Solution**

Engineered a Linear Projection Head to align NTV3 (1536d) and ESM-2 (1280d).

HIGH- THROUGHPUT INDEXING

- **Solution**

Built an Asynchronous Batch Embedding API that successfully indexed 35,000+ DNA and 25,000+ Protein sequences

CROSS - SPECIES GENERALIZATION

- **Solution**

Validated robustness on Fruit Fly (*Drosophila*) data, achieving 28.8% accuracy without specific fine-tuning. Used Contrastive Loss to maximize learning efficiency from limited pairs.

CHALLENGES & SOLUTIONS



06

IMPACT & VALUE

FOR RESEARCHERS

- A "Google" for biology .
- Search by sequence, not keywords

FOR CLINICIAN

- Zero-Shot Variant Scoring provides instant second opinions on rare genetic mutations (VUS).

FOR PHARMA

- The RL-Ready Reward API acts as a fitness function for Generative AI, accelerating drug candidate selection.

SCALABILITY

- The Batch API allows processing entire cohorts of patients in seconds, not days.

IMPACT & VALUE

THANK YOU

THE FUTURE OF BIOLOGY IS GEOMETRIC

QR code for
GitHub
repository

